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Math 31 — Differential Equations
Section 7.3.1

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3. $\mathcal{L}\{t^3e^{-2t}\}$

Shift s by $(s-(-2))$:
$\mathcal{L}\{t^3e^{-2t}\} = \frac{6}{(s+2)^3}$

7. $\mathcal{L}\{e^t\sin(3t)\}$

Shift by $(s-1)$:
$\mathcal{L}\{e^t\sin(3t)\} = \frac{3}{(s-1)^2+9}$

9. $\mathcal{L}\{(1-e^t+3e^{-4t})\cos(5t)\}$

Distributing the $\cos(5t)$ into each term:
$\mathcal{L}\{\cos(5t) - e^t\cos(5t) + 3e^{-4t}\cos(5t)\}$
Which gives us:
$\mathcal{L}\{\cos(5t) - e^t\cos(5t) + 3e^{-4t}\cos(5t)\}$
Taking the Laplace:
$\frac{s}{s^2+25} - \frac{(s-1)}{(s-1)^2+25} + \frac{3s}{(s+4)^2+25}$

11. $\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^3}\right\}$

The s term here is shifted by 3 and has a multiplicity of 3:
Getting the inverse into form:
$\frac{1}{2!}\mathcal{L}^{-1}\left\{\frac{2!}{(s+3)^{2+1}}\right\}$
Which gives us our final answer of:
$\boldsymbol{\left[\frac{t^2e^{-3t}}{2}\right]}$

13. $\mathcal{L}^{-1}\left\{\frac{1}{s^2-6s+10}\right\}$

Check if the denominator is reducible:
$D = \sqrt{(-6)^2 - (4\cdot 10\cdot 1)} \rightarrow D = \sqrt{-4}$
$\therefore$ Irreducible:
Utilize completing the square:
$\mathcal{L}^{-1}\left\{\frac{1}{s^2-6s+9-9+10}\right\} \rightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2+1}\right\}$
Which gives us our final answer of:
$\boldsymbol{[e^{3t}\sin(t)]}$

15. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+5}\right\}$

First complete the square on the bottom:
$\frac{s}{(s+2)^2+1}$
Take inverse which gives us our final answer:
$\boldsymbol{[e^{-2t}\cos(t)]}$

16. $\mathcal{L}^{-1}\left\{\frac{2s+5}{s^2+6s+34}\right\}$

Complete the square on the denominator:
$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s+3)^2+25}\right\}$
Separate the fraction:
$\mathcal{L}^{-1}\left\{\frac{2s}{(s+3)^2+25} + \frac{5}{(s+3)^2+25}\right\}$

Which results in:

$$\boldsymbol{[2e^{-3t}\cos(5t) + 5e^{-3t}\sin(5t)]}$$

17. $\mathcal{L}^{-1}\left\{\frac{s}{(s+1)^2}\right\}$

Setup the PFD:

$$\frac{s}{(s+1)^2} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2}$$

Multiplying both sides by $(s+1)^2$ results in:

$$s = As + A + B$$

And by comparing coefficents we get that:

$$A = 1$$

$$A + B = -1$$

$$\therefore$$

$$A = 1$$

$$B = -1$$

Our PFD then becomes:

$$\frac{1}{(s+1)} - \frac{1}{(s+1)^2}$$

And thus the inverse Laplace is:

$$\boldsymbol{[e^{-t} - te^{-t}]}$$

18. $\mathcal{L}^{-1}\left\{\frac{5s}{(s-2)^2}\right\}$

Start PFD:

$$\frac{5s}{(s-2)^2} = \frac{A}{(s-2)} + \frac{B}{(s-2)^2}$$

Multiply both sides by $(s-2)^2$

$$5s = As - 2A + B$$

Comparing coefficents:

$$A = 5$$

$$-2A + B = 0$$

$$\therefore$$

$$A = 5$$

$$B = 10$$

Resulting in:

$$\frac{5}{(s-2)} + \frac{10}{(s-2)^2}$$

And after taking our inverse Laplace our final answer is:

$$\boldsymbol{[5e^{2t} + 10te(2t)]}$$

19. $\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2(s+1)^3}\right\}$

PFD:

$$\frac{2s-1}{s^2(s+1)^3} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{(s+1)^2} + \frac{E}{(s+1)^3}$$

Multiplying both sides by $s^2(s+1)^3$

$$2s - 1 = As^4 + 3As^3 + 3As^2 + As + Bs^3 + 3Bs^2 + 3Bs + B + Cs^4 + 2Cs^3 + Cs^2 + Ds^3 + Ds^2 + Es^2$$

Setting up our augmented matrix:

$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 0 & 0 \\ 3 & 1 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 1 & 1 & 0 \\ 1 & 3 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 & -1 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & -5 \\ 0 & 0 & 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & -3 \end{array}\right)$$

Which give us our decomposed fraction:

$$\frac{5}{s} - \frac{1}{s^2} - \frac{5}{s+1} - \frac{4}{(s+1)^2} - \frac{3}{(s+1)^3}$$

Taking the inverse Laplace gives us the final answer of:

$$\left[\mathbf{5 - t - 5e^{-t} - 4te^{-t} - \tfrac{3}{2}t^2e^{-t}}\right]$$

27. $y'' + 2y' + y = 0 \mid y(0) = 1, y'(0) = 1$

Take the Laplace of both sides:

$$s^2Y(s) - sy(0) - y'(0) + 2sY(s) - 2y(0) + Y(s) = 0$$

Substitute the initial conditions:

$$s^2Y(s) - s - 1 + 2sY(s) - 2 + Y(s) = 0$$

Isolating for $Y(s)$

$$Y(s)(s^2 + 2s + 1) = s + 3$$

Divide by $(s^2 + 2s + 1)$

$$Y(s) = \frac{s+3}{(s+1)^2}$$

Decomposing our fraction:

$$\frac{s+3}{(s+1)^2} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2}$$

Multiplying by $(s+1)^2$ gives us:

$$s + 3 = As + A + B$$

Comparing coefficents:

$$A = 1$$

$$A + B = 3$$

$$\therefore$$

$$A = 1$$

$$B = 2$$

Resulting fraction:

$$Y(s) = \frac{1}{s+1} + \frac{2}{(s+1)^2}$$

Taking the inverse Laplace gives us our solution:

$$\left[\mathbf{y = e^{-t} + 2te^{-t}}\right]$$

29. $y'' - 6y' + 9y = t \mid y(0) = 0, y'(0) = 1$

Take the Laplace of both sides:

$$s^2Y(s) - sy(0) - y'(0) - 6sY(s) + 6y(0) + 9Y(s) = \frac{1}{s^2}$$

Substitute in our initial conditions:

$$s^2Y(s) - 1 - 6sY(s) + 9Y(s) = \frac{1}{s^2}$$

Add over the 1 and isolate for $Y(s)$

$$Y(s)(s^2 - 6s + 9) = \frac{1+s^2}{s^2}$$

Isolating for $Y(s)$

$$Y(s) = \frac{1+s^2}{s^2(s-3)^2}$$

Decomposing our fraction:

$$\frac{1+s^2}{s^2(s-3)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-3)} + \frac{D}{(s-3)^2}$$

Multiplying both sides by $s^2(s-3)^3$

$$1 + s^2 = As^3 - 6As^2 + 9As + Bs^2 - 6Bs + 9B + Cs^3 - 3Cs^2 + Ds^2$$

Setting up our matrix:

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ -6 & 1 & -3 & 1 & 1 \\ 9 & -6 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 & 1 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{2}{27} \\ 0 & 1 & 0 & 0 & \frac{1}{9} \\ 0 & 0 & 1 & 0 & -\frac{2}{27} \\ 0 & 0 & 0 & 1 & \frac{10}{9} \end{array}\right)$$

Which gives us our decomposed fraction of:

$$Y(s) = \frac{2}{27s} + \frac{1}{9s^2} - \frac{2}{27(s-3)} + \frac{10}{(s-3)^2}$$

$\mathcal{L}^{-1}$ gives us our solution:

$$\left[\mathbf{y = \tfrac{2}{27} + \tfrac{t}{9} - \tfrac{2}{27}e^{3t} + 10te^{-3t}}\right]$$

32. $2y'' + 20y' + 51y = 0 \mid y(0) = 2, y'(0) = 0$

Laplace both sides:

$$2s^2Y(s) - 2sy(0) - 2y'(0) + 20sY(s) - 20y(0) + 51Y(s) = 0$$

Substituing in our initial conditions:

$$2s^2Y(s) - 4s + 20sY(s) - 40 + 51Y(s) = 0$$

Isolate for $Y(s)$:

$$Y(s) = \frac{4s+40}{2s^2+20s+51}$$

Factor out a 2 on the denominator and numerator:

$$Y(s) = \frac{(2)(2s+20)}{(2)(s^2+10s+\frac{51}{2})}$$

Cancel the 2s which gives us a simplified fraction:

$$\frac{2s+20}{s^2+10s+\frac{51}{2}}$$

Complete the square:

$$\frac{2s+20}{s^2+10s+25-25+\frac{51}{2}} \longrightarrow \frac{2s+10+10}{(s+5)^2+\frac{1}{2}}$$

Split the fraction:

$$\frac{2(s+5)}{(s+5)^2+\frac{1}{2}} + \frac{10}{(s+5)^2+\frac{1}{2}}$$

Get into form:

$$\frac{2(s+5)}{(s+5)^2+\sqrt{\frac{1}{2}}^2} + \frac{10}{(s+5)^2+\sqrt{\frac{1}{2}}^2}$$

Take $\mathcal{L}^{-1}$

$$y = 2e^{-5t}\cos\left(\frac{t}{\sqrt{2}}\right) + 10\sqrt{2}e^{-5t}\sin\left(\frac{t}{\sqrt{2}}\right)$$

34. $y'' - 2y' + 5y = 1 + t \mid y(0) = 0, y'(0) = 4$

Take $\mathcal{L}$:

$$s^2Y(s) - sy(0) - y'(0) - 2sY(s) - 2y(0) + 5Y(s) = \frac{s+1}{s^2}$$

Substituting in our values:

$$s^2Y(s) - 4 - 2sY(s) + 5Y(s) = \frac{s+1}{s^2}$$

Solving for $Y(s)$

$$Y(s) = \frac{4}{s^2-2s+5} + \frac{s+1}{s^2(s^2-2s+5)}$$

Completing the square:

$$Y(s) = \frac{4}{(s-1)^2+4} + \frac{s+1}{s^2((s-1)^2+4)}$$

We must decompose the right fraction:

$$\frac{s+1}{s^2((s-1)^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{(s-1)^2+4}$$

Multiply both sides by $s^2((s-1)^2+4)$:

$$s + 1 = As^3 - 2As^2 + 5As + Bs^2 - 2Bs + 5B + Cs^3 + Ds^2$$

Coefficient matrix:

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 \\ 5 & -2 & 0 & 0 & 1 \\ 0 & 5 & 0 & 0 & 1 \end{array}\right) \longrightarrow \text{RREF: } \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{7}{25} \\ 0 & 1 & 0 & 0 & \frac{1}{5} \\ 0 & 0 & 1 & 0 & -\frac{7}{25} \\ 0 & 0 & 0 & 1 & \frac{9}{25} \end{array}\right)$$

Backsubstituting:

$$\frac{s+1}{s^2((s-1)^2+4)} = \frac{7}{25s} + \frac{1}{5s^2} + \frac{-7s+9}{25((s-1)^2+4)}$$

Going back to our equation we get:

$$\frac{4}{(s-1)^2+4} + \frac{7}{25s} + \frac{1}{5s^2} + \frac{-7s+9}{25((s-1)^2+4)}$$

Getting the equation into form for inverse Laplace:

$$\frac{4}{(s-1)^2+4} + \frac{7}{25s} + \frac{1}{5s^2} + \frac{-7}{25}\left(\frac{s-63}{(s-1)^2+4}\right)$$

Simplifying further:

$$\frac{4}{(s-1)^2+4} + \frac{7}{25s} + \frac{1}{5s^2} + \frac{-7}{25}\left(\frac{(s-1)}{(s-1)^2+4} - \frac{62}{(s-1)^2+4}\right)$$

Taking $\mathcal{L}^{-1}$

$$y = 2e^t \sin(2t) + \frac{7}{25} + \frac{1}{5}t - \frac{7}{25}(e^t \cos(2t) - 31e^t \sin(2t))$$